

Effect of fibre, antispasmodics, and peppermint oil in the treatment of irritable bowel syndrome: systematic review and meta-analysis

Objective To determine the effect of fibre, antispasmodics, and peppermint oil in the treatment of irritable bowel syndrome. **Design** Systematic review and meta-analysis of randomised controlled trials. **Data sources** Medline, Embase, and the Cochrane controlled trials register up to April 2008. **Review methods** Randomised controlled trials comparing fibre, antispasmodics, and peppermint oil with placebo or no treatment in adults with irritable bowel syndrome were eligible for inclusion. The minimum duration of therapy considered was one week, and studies had to report either a global assessment of cure or improvement in symptoms, or cure of or improvement in abdominal pain, after treatment. A random effects model was used to pool data on symptoms, and the effect of therapy compared with placebo or no treatment was reported as the relative risk (95% confidence interval) of symptoms persisting. **Results** 12 studies compared fibre with placebo or no treatment in 591 patients (relative risk of persistent symptoms 0.87, 95% confidence interval 0.76 to 1.00). This effect was limited to ispaghula (0.78, 0.63 to 0.96). Twenty two trials compared antispasmodics with placebo in 1778 patients (0.68, 0.57 to 0.81). Various antispasmodics were studied, but otilonium (four trials, 435 patients, relative risk of persistent symptoms 0.55, 0.31 to 0.97) and hyoscine (three trials, 426 patients, 0.63, 0.51 to 0.78) showed consistent evidence of efficacy. Four trials compared peppermint oil with placebo in 392 patients (0.43, 0.32 to 0.59). **Conclusion** Fibre, antispasmodics, and peppermint oil were all more effective than placebo in the treatment of irritable bowel syndrome.